

SAT Unit 4.3

Trigonometry

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Overview

- 1 Trigonometric Ratios of Acute Angles
- 2 Radian Measure
- 3 Trigonometric In Unit Circle

Trigonometric Ratios of Acute Angles

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Trigonometric Ratios and Cofunctions

Trigonometric Ratios: A ratio of the lengths of sides of a right triangle.

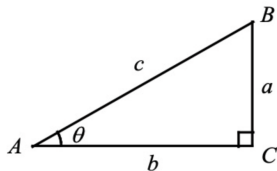
$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{a}{c}, \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{b}{c}, \quad \tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{a}{b}$$

Cofunction Identity: In a right triangle ABC , angles A and B are complementary:

$$m\angle A + m\angle B = 90^\circ$$

So:

$$\sin(\theta) = \cos(90^\circ - \theta), \quad \tan(\theta) = \cot(90^\circ - \theta)$$



Trigonometric Ratios Are Constant Ratios

Key Concept: Trigonometric functions like $\sin \theta$, $\cos \theta$, and $\tan \theta$ represent **ratios of side lengths** in a right triangle.

Important Insight: For any given angle θ , the ratio is **always the same** — no matter the size of the triangle — as long as it's a right triangle.

Examples:

- $\sin(30^\circ) = \frac{1}{2}$ — In **every** right triangle with a 30° angle, the opposite side is half the hypotenuse.
- $\cos(60^\circ) = \frac{1}{2}$ — In any right triangle with a 60° angle, the adjacent side is half the hypotenuse.
- $\tan(45^\circ) = 1$ — In any right triangle with a 45° angle, the opposite and adjacent sides are equal.

SAT Tip: Memorize the sine, cosine, and tangent of special angles: 30° , 45° , and 60° — they appear frequently on the test!

Question

$$RS = 20$$

$$ST = 48$$

$$TR = 52$$

The side lengths of right triangle RST are given. Triangle RST is similar to triangle UVW , where S corresponds to V and T corresponds to W . What is the value of $\tan W$?

A. $\frac{5}{13}$

B. $\frac{5}{12}$

C. $\frac{12}{13}$

D. $\frac{12}{5}$

Answer

Since triangle RST is a right triangle with hypotenuse $TR = 52$, and legs $RS = 20$, $ST = 48$, the angle at W corresponds to angle T .

$$\tan(W) = \frac{\text{opposite}}{\text{adjacent}} = \frac{RS}{ST} = \frac{20}{48} = \frac{5}{12}$$

Answer: B. $\frac{5}{12}$

Question

Triangle ABC is similar to triangle DEF , where A corresponds to D and C corresponds to F . Angles C and F are right angles. If $\tan(A) = \sqrt{3}$ and $DF = 125$, what is the length of $\overline{DE}$?

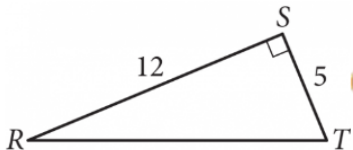
- A. $125 \cdot \frac{\sqrt{3}}{3}$
- B. $125 \cdot \frac{\sqrt{3}}{2}$
- C. $125\sqrt{3}$
- D. 250

Answer

Answer:

D. 250

Question: Trigonometric Expression



In triangle RST above, point W (not shown) lies on $\overline{RT}$. What is the value of

$$\cos(\angle RSW) - \sin(\angle WST)?$$

Answer Explanation: Trigonometric Expression in Triangle

Using the **Cofunction Identity**

Final Answer:

Radian Measure

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Understanding Radian Measure

Definition of Radian: A radian is the angle created when the arc length equals the radius.

$$1 \text{ radian} = \frac{s}{r}$$

- s : Arc length
- r : Radius

Important Note: There are 2π radians in a full circle

Degree to Radian Conversion:

$$\theta \text{ (in radians)} = \theta^\circ \cdot \frac{\pi}{180}$$

Radian Conversion to Degree to Conversion:

$$\theta \text{ (in degree)} = \theta \cdot \frac{180}{\pi}$$

Trigonometric In Unit Circle

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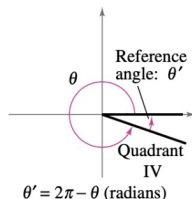
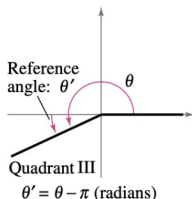
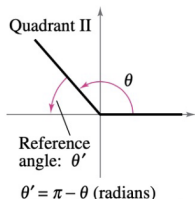
What If the Angle is Greater Than 90 Degrees?

Reference Angle: The reference angle is the **acute angle** (less than 90°) formed between the terminal side of an angle in standard position and the x -axis.

Why Use It?

- Trig values for any angle θ (in degrees or radians) can be found by using its reference angle.
- The sine, cosine, and tangent of the original angle have the same absolute value as the reference angle — with signs determined by the quadrant.

How to Find the Reference Angle:



What If the Angle is Greater Than 90 Degrees?

Big idea: for angles outside a right triangle, do not start by drawing a new triangle. Start with the angle's position on the coordinate plane.

Use this 4-step routine:

- 1 Draw the angle in **standard position**: start on the positive x -axis and rotate counterclockwise.
- 2 Identify the **quadrant** where the terminal side lands.
- 3 Find the **reference angle**: the acute angle between the terminal side and the x -axis.
- 4 Use the reference-angle value, then attach the correct sign.

Quadrant	Positive trig ratio
I	$\sin, \cos, \tan$
II	$\sin$
III	$\tan$

Example: Angle Greater Than 90 Degrees

Find $\sin(150^\circ)$.

- 1 150° lands in **Quadrant II**.
- 2 The reference angle is $180^\circ - 150^\circ = 30^\circ$.
- 3 $\sin(30^\circ) = \frac{1}{2}$.
- 4 Sine is positive in Quadrant II.

$$\sin(150^\circ) = \frac{1}{2}$$

Trigonometric Functions on the Unit Circle

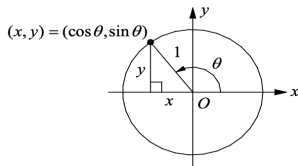
What the unit circle gives us:

Put angle θ in standard position and look at the point where the terminal side hits the circle. If that point is (x, y) , then:

$$\cos(\theta) = x, \quad \sin(\theta) = y, \quad \tan(\theta) = \frac{y}{x} \quad (x \neq 0)$$

So when solving a unit-circle problem:

- 1 Reduce the angle to a familiar terminal position.
- 2 Find the point (x, y) on the unit circle.
- 3 Read sin from y , read cos from x , and use y/x for tan.



Question: Trigonometric Constant Expression

The value of $\tan\left(\frac{49\pi}{6}\right)$ can be written as $k\sqrt{3}$, where k is a constant. What is the value of k ?

Choices:

- A. $\frac{1}{3}$
- B. $\frac{1}{\sqrt{3}}$
- C. 1
- D. 3

Answer Explanation: Trigonometric Constant Expression

To evaluate $\tan\left(\frac{49\pi}{6}\right)$, reduce the angle:

$$\frac{49\pi}{6} = 8\pi + \frac{\pi}{6}$$

Since 2π is one full rotation, this is coterminal with $\frac{\pi}{6}$. So,

$$\tan\left(\frac{49\pi}{6}\right) = \tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}}$$

Thus, we write:

$$\tan\left(\frac{49\pi}{6}\right) = \frac{1}{\sqrt{3}} = k\sqrt{3} \Rightarrow k = \frac{1}{3}$$

Correct Answer: A. $\frac{1}{3}$

Unit Circle - Find Coordinate

Point N lies on a unit circle in the xy -plane and has coordinates $(1, 0)$. Point O is the center of the circle and has coordinates $(0, 0)$. Point M also lies on the unit circle, and the measure of angle $\angle NOM$ is $\frac{266\pi}{3}$ radians. If the coordinates of point M are (a, b) , where a and b are constants, what is the value of b ?

- A. $\frac{1}{2}$
- B. $\frac{\sqrt{3}}{2}$
- C. $-\frac{1}{2}$
- D. $-\frac{\sqrt{3}}{2}$

Answer to Unit Circle Problem

Since the angle is 9π , which is equivalent to π (modulo 2π), the terminal point is the same as π on the unit circle.

$$\cos(\pi) = -1, \quad \sin(\pi) = 0 \quad \Rightarrow \quad M = (-1, 0)$$

But based on options that suggest a reference angle of $\frac{\pi}{6}$, observe:

$$\frac{27\pi}{3} = 9\pi = 4(2\pi) + \pi \Rightarrow \text{same terminal side as } \pi$$

This implies a point with coordinate:

$$a = -1, \text{ not among options.}$$

However, if the actual question is about angle $\frac{2\pi}{3}$ (possibly a typo), then:

$$\sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

Correct Answer: B.



Practices

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Triangle Side Length

Question: In triangle ABC , the measure of angle A is 25° , angle C is a right angle, and the length of BC is 16. Which of the following represents the length of AC ?

A $16 \sin 25^\circ$

B $16 \tan 25^\circ$

C $\frac{16}{\sin 25^\circ}$

D $\frac{16}{\tan 25^\circ}$

Answer Choices

Answer: D

Trigonometry

Question: Triangle JKL is similar to triangle RST , where angle J corresponds to angle R and angles K and S are right angles. If $\cos(L) = \frac{119}{169}$, what is the value of $\tan(T)$?

- A $\frac{120}{169}$
- B $\frac{119}{120}$
- C $\frac{120}{119}$
- D $\frac{169}{119}$

Solution

Solution: Since the triangles JKL and RST are similar, corresponding angles are congruent, so L corresponds to T . Given:

$$\cos(L) = \frac{119}{169}$$

Using the Pythagorean identity:

$$\sin^2(L) + \cos^2(L) = 1 \implies \sin(L) = \sqrt{1 - \left(\frac{119}{169}\right)^2}$$

$$\tan(L) = \frac{\sin(L)}{\cos(L)} = \frac{\sqrt{1 - \left(\frac{119}{169}\right)^2}}{\frac{119}{169}} = \frac{\sqrt{169^2 - 119^2}}{119}$$

$$169^2 - 119^2 = (169 - 119)(169 + 119) = 50 \times 288 = 14400$$
$$\sqrt{14400} = 120$$

So:

$$\tan(T) = \tan(L) = \frac{120}{119}$$

Option C: $\frac{120}{119}$

Fraction of Circumference

Question: Point O is the center of a circle. The measure of arc MN on this circle is $\frac{4\pi}{3}$ radians. The length of arc MN is what fraction of the circumference of the circle?

Solution

Solution: - Recall that the full circumference of a circle is 2π radians in angle measure (equivalent to 360°). - The measure of arc MN is given as $\frac{4\pi}{3}$ radians. - To find the fraction of the circumference represented by this arc:

$$\text{Fraction} = \frac{\frac{4\pi}{3}}{2\pi} = \frac{4}{6} = \frac{2}{3}$$

$$\boxed{\frac{2}{3}}$$

Acute Angle with the x-axis

Question: In the xy -plane, segment OA has endpoints $(0,0)$ and $(\sqrt{3}, \sqrt{3})$. What is the measure, in radians, of the acute angle formed by segment OA and the line $y = 0$?

- A) $\frac{\pi}{6}$
- B) $\frac{\pi}{4}$
- C) $\frac{\pi}{3}$
- D) $\frac{\pi}{2}$

Solution

Solution: - The segment OA has a slope:

$$\tan(\theta) = \frac{\sqrt{3}}{\sqrt{3}} = 1$$

- Therefore:

$$\theta = \arctan(1) = \frac{\pi}{4}$$

- Thus:

$$\boxed{\frac{\pi}{4}}$$

Length of Side in Similar Triangles

Question: Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angles B and E are right angles. If $AB = 2$, $DF = 12\sqrt{2}$, and $\tan(A) = \sqrt{7}$, what is the length of $\overline{DE}$?

Solution

Solution:

6

Thank You!

We appreciate your time and focus.

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